Paper 7.25 - Toolkit for the Discrete-Continuous Bridge

CQE-paper-07.25
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-07.25.md

Paper 7.25 - Toolkit for the Discrete-Continuous Bridge

Purpose

Paper 7.25 describes the tools for reviewing the discrete-continuous bridge. These tools expose sample-preserving interpolation and receipt retention; they do not prove between-sample dynamics.

Review Objects

The toolkit works with:

```
indexed trace       = [(0,x0), ..., (n,xn)]
bridge function     = piecewise-linear F
sample error        = abs(F(k)-xk)
endpoint agreement  = adjacent segments share samples
discrete receipt    = original proof or trace record
```

Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-07/verify_discrete_continuous_bridge.py
production/formal-papers/CQE-paper-07/discrete_continuous_bridge_receipt.json
```

Additional source and kernel files:

```
production/papers/CQE-paper-07/SOURCE.md
production/papers/CQE-paper-07/01-CQE-formal/FORMAL.md
production/papers/CQE-paper-07/02-CQE-tool/TOOL.md
production/papers/CQE-paper-07/03-CQE-workbook/WORKBOOK.md
production/paper-kernels/papers/CQE-paper-07/PAPER_KERNEL.md
```

The kernel notes currently mark `production/papers/CQE-paper-07/02-CQE-tool/run.py` as missing, so the promoted verifier is the replayable route for this paper.

Analog Toolkit

A physical reconstruction requires:

- graph paper or any coordinate surface,
- a finite list of indexed values,
- a straightedge or equivalent mark,
- a receipt line.

Procedure:

```
plot each indexed sample
connect adjacent samples with straight segments
check that every original sample remains on the line
write the sample error for each index
mark between-sample interpretation as unproven unless separately verified
keep the original discrete receipt beside the drawing
```

Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

```
What is the interpolation error at integer samples?
Do adjacent segments agree at shared endpoints?
Does sample preservation prove between-sample physics?
What must stay attached to the drawing?
```

Expected answers:

```
zero
yes
no
the original discrete receipt
```

Boundary

Paper 7.25 is a toolkit supplement. If a later trace claims physical dynamics between samples, it must pass Paper 7.50's claim contract and attach an additional theorem.